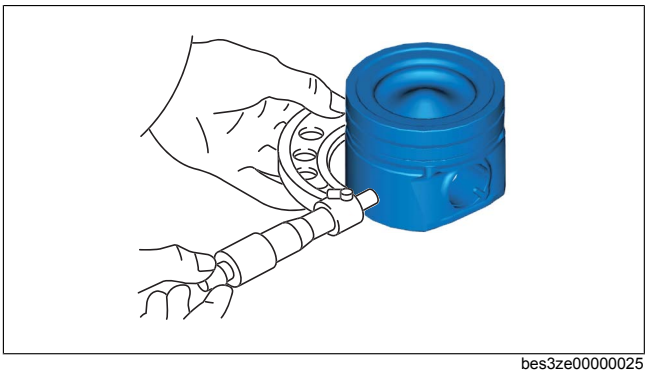
PISTON INSPECTION

id011000502100

Caution
- **If the piston is replaced, replace the piston, piston pin, and the snap ring as a single component.**

1. Measure the piston outer diameter using the micrometer. Perform the inspection at the following location.
   - **10.0 mm {0.394 in}** from the lower end of the piston (area with no coating on piston skirt)
     — If it is not within the specification, replace the piston.


bes3ze00000025

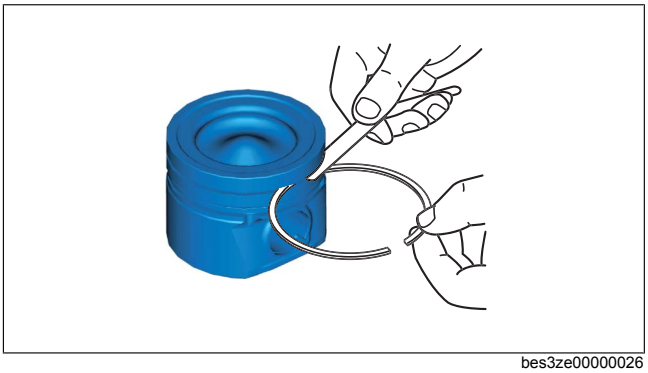
Standard piston outer diameter
75.958—75.984 mm {2.9905—2.9914 in}

2. Measure the cylinder bore diameter. (See CYLINDER BLOCK INSPECTION.)
3. Calculate the cylinder-to-piston clearance from the cylinder bore diameter and the piston outer diameter.
   - If the clearance exceeds the maximum specification, replace the piston or cylinder block.

Standard clearance between piston and cylinder
0.031—0.049 mm {0.0013—0.0019 in}

Maximum clearance between piston and cylinder
0.067 mm {0.0026 in}

4. Measure the piston-to-ring groove clearance along the perimeter using a feeler gauge. For the O-ring, measure the clearance with the O-ring assembled to the piston.
   - If the clearance exceeds the maximum specification, replace the piston or piston ring.


bes3ze00000026

Standard clearance between piston ring and ring groove
Top: 0.10—0.14 mm {0.0040—0.0055 in}
Second: 0.05—0.09 mm {0.002—0.003 in}
Oil: 0.05—0.09 mm {0.002—0.003 in}

Maximum clearance between piston ring and ring groove
Top: 0.16 mm {0.0063 in}
Second: 0.12 mm {0.0047 in}
Oil: 0.11 mm {0.0043 in}